Lesson 14B Pascal, Archimedes, and Bernoulli

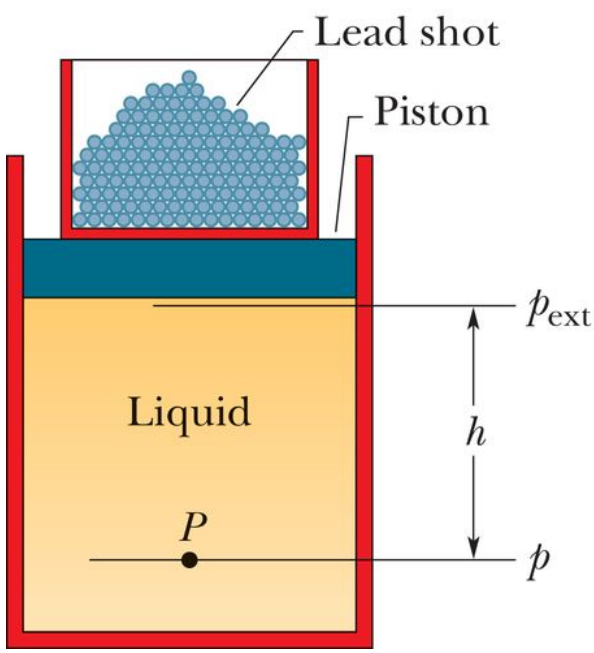
Pascal's Principle

Pascal's Principle describes how pressure is transmitted in a confined, incompressible fluid:

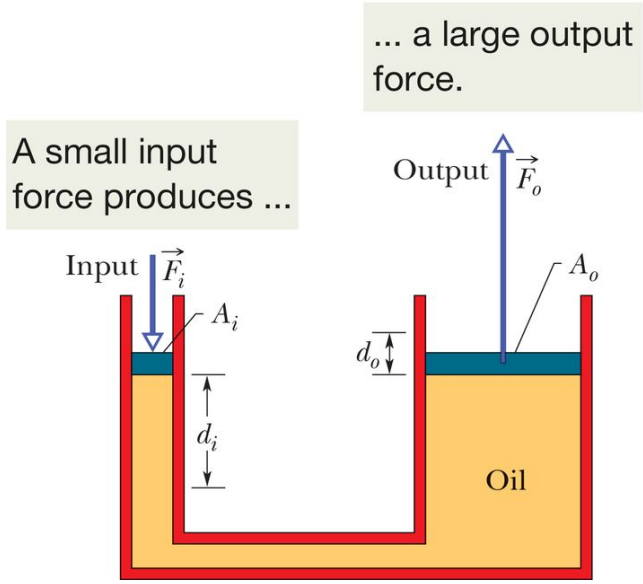
When an external pressure is applied to a fluid in a closed container, that pressure change is transmitted equally and undiminished throughout the entire fluid.

Consider a cylinder filled with an incompressible fluid:

- Applying an external pressure p_{ext} causes the pressure at every point within the fluid to increase by the same amount, regardless of depth.



This principle is the basis for **hydraulic systems**, such as the **hydraulic lever**:



- If an input force F_i is applied to a piston with area A_i, it produces a pressure:

$$p = \frac{F_\mathrm{i}}{A_\mathrm{i}}$$

- This pressure is transmitted throughout the fluid and results in an output force F_o on a second piston of area A_o:

$$\frac{F_\mathrm{i}}{A_\mathrm{i}} = \frac{F_\mathrm{o}}{A_\mathrm{o}}$$

- The input and output displacements, d_i and d_o, are related such that the work done is conserved:

$$F_\mathrm{i} d_\mathrm{i} = F_\mathrm{o} d_\mathrm{o}$$

Mechanical Advantage: A small input force F_i applied over a longer distance can produce a much larger output force F_o, over a shorter distance, by using a larger output piston area A_o.

Archimedes' Principle

When an object is submerged in a fluid, it experiences an upward force due to pressure differences between the top and bottom of the object. This upward force is called the **buoyant**

force, denoted F_{b}.

Archimedes' Principle: The buoyant force

F_{b} acting on a submerged object is equal to the weight of the fluid displaced by the object.

$$F_{\mathrm{b}} = \rho_{\mathrm{f}} V g$$

Where:

- ρ_{f} is the density of the fluid
- V is the volume of displaced fluid
- g is the acceleration due to gravity

Floating Objects

When an object is floating in equilibrium:

- The buoyant force equals the object's weight:

$$F_{\mathrm{b}} = mg$$

- The object displaces just enough fluid to support its own weight.

Apparent Weight in a Fluid

The **apparent weight** of an object submerged in a fluid is less than its actual weight due to the upward buoyant force:

$$\text{Apparent weight} = \text{Actual weight} - F_{\mathrm{b}}$$

Or algebraically:

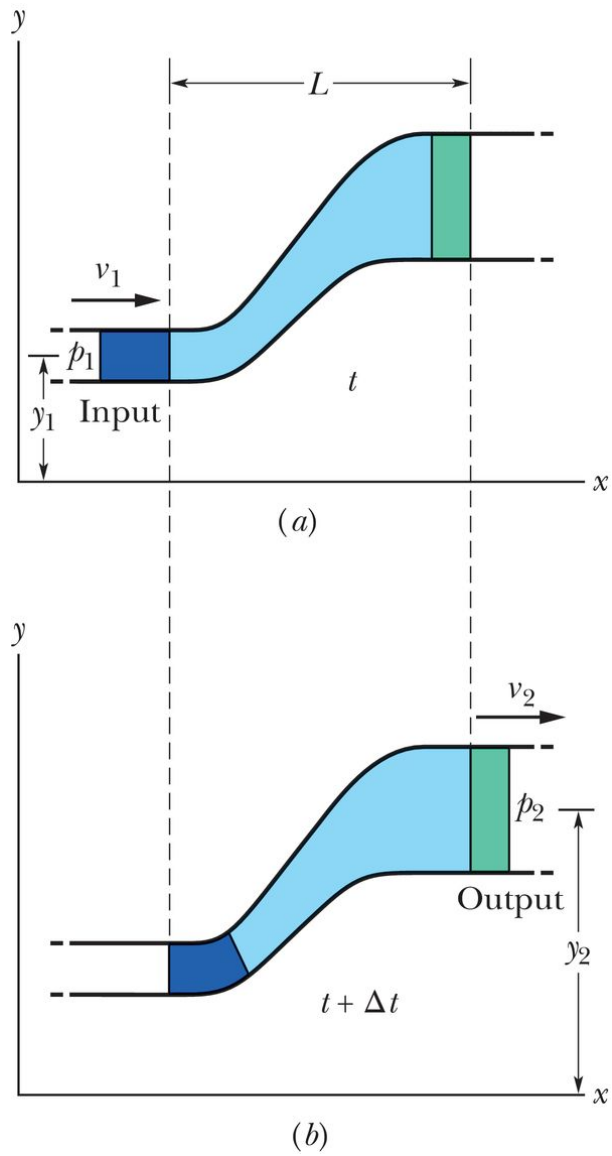
$$F_{\text{apparent}} = mg - F_{\mathrm{b}}$$

How taking a bath led to Archimedes' principle - Mark Salata

Bernoulli's Equation

Bernoulli's Equation describes the conservation of energy in a flowing fluid. It's based on the principle that the total mechanical energy per volume remains constant along a streamline in an ideal (non-viscous, incompressible) fluid.

Consider fluid flowing through a pipe with changing height and speed, as shown below:



$$p + \frac{1}{2}\rho v^2 + \rho g y = \text{constant}$$

- pp: pressure energy density
- $\frac{1}{2}\rho v^2$: kinetic energy density
- $\rho g y$ρgy: gravitational potential energy density

This equation tells us how pressure, speed, and height are related in a moving fluid.

Special Cases:

- **Fluid at Rest**: $v = 0$, so Bernoulli's Equation reduces to the hydrostatic pressure formula:

$$p = p_0 + \rho g h$$

- **Horizontal Flow** (constant height): $y_1 = y_2$, so it simplifies to:

$$p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2$$

This form explains why fluids **speed up in narrow pipes** and why **pressure drops in faster flows**—like in airplane wings or venturi tubes.